

Deep Learning

Exercise 13: Exam Preparation

Room: **BIN-1-B.01**

Instructor: Manuel Günther

Email: guenther@ifi.uzh.ch

Office: AND 2.54

Friday, May 31, 2024

Outline

- 1 Review
- 2 Details of the Examination

Review of the Exercise

Ex. 1. Python Programming

- Basic introduction to `python`

Ex. 2. Two-layer Networks

- Introduction `numpy`, `matplotlib`
 - Elementwise operations
 - Matrix multiplication
 - Plotting code provided in exam

Ex. 3. Data Science Pipeline

- Data loading via `pandas`
- Traditional ML with `sklearn`

Assignment 1

- Two-layer network in `numpy`
- Handling of bias neurons
- Manual gradient computation
- Gradient implementation
- Iterative gradient descent
- Function approximation
- Manual Weight decay

Review of the Exercise

Assignment 2

- Derivative of stable SoftMax
- Implementing SoftMax in `numpy`
- Dataset loading in `numpy`
- Input data normalization
- Binary classification
- Categorical classification
- Accuracy computation

Ex. 5: PyTorch

- `torch.tensor` data structure
- Building a network
 - `Sequential`, custom `Modules`
 - Different layer types
- `torchvision` datasets
 - Input transforms, `DataLoaders`
 - Custom `Datasets`
- Loss functions, Optimizers
- Basic training loop on GPU
- `torch.autograd.Function`

Review of the Exercise

Assignment 3

- Receptive field computation
- Analyzing loss behavior
- Convolutional layers
- Pooling and striding
- Flattening
- Fully-connected layers
- Parameter freezing and fine-tuning
- Confusion matrix plotting

Ex. 7: PyTorch Pipeline

- Data loading and analysis
- Training, validation and test
- Custom `Dataset` implementation
- Loss functions, optimizers
- Learning rate schedulers
- Training and validation loop

Review of the Exercise

Assignment 4

- PCA vs. two-layer AE
- Entropic open-set loss
- Convolutional auto-encoder
- Out-of-distribution detection
- Sample reconstruction
- Open-set recognition
- Class centroids

Ex. 9: Generative Networks

- Data transforms
- Generator network
- Discriminator network
- Alternating training
- Conditional GAN

Review of the Exercise

Assignment 5

- Feature map dimensions
- Time series processing
- 1D convolutional network
- RNN and LSTM
- Self-attention implementation
- Bias mitigation
 - Class weighting in CCE

Ex. 11: Explainable AI

- Pre-trained network
 - Function overloading
- Network with multiple outputs
- Weight matrix access
- Class activation mapping
- Back-propagation to input

Ex. 12: Foundation Models

- Comparison of embeddings
 - Zero-shot classification
- Chaining of foundation models

Details of the Examination

Examination

- Friday June 13, 2025, 8:00 - 10:00
- Onsite (**Y15-G-20 / Y15-G-40**)
- Get seated 15 minutes before
 - Bring your Student ID card
- Several assignments
- Several tasks per assignment
 - Theoretical and practical
- Jupyter notebook per assignment
 - Several tasks in one notebook
- 2 hours time (might be short)

Examination Type

- Open-book exam
- Any resources allowed
 - Except for human resources
 - Except for generative AI

Test Run (Mock Exam)

- Participate in the test run
- Available 6.6.2025
- Test download and upload
- Installation instruction

Details of the Examination

Choice Questions

- True/false questions
 - 1 point for correct answer
 - **-0.5 points for incorrect answer**
- Multiple choice questions
 - 1 point for correct answer
 - A given number of answers must be selected
 - Deselect one answer to choose a different one

Details of the Examination

Free Text Boxes

- Free text boxes for theory
 - Write short text answers
- Symbols and LaTeX enabled
 - But not required
- Write simple text:
 - Use df/dx to indicate $\frac{\partial f}{\partial x}$
 - Use a^{b} to indicate a^b
 - Use a_b to indicate a_b
 - Use parentheses $() [] \{\}$
- Anything readable is accepted

Coding

- Single-choice question
 - Structural element
 - Not required to click
 - Points for sub-tasks
- File upload boxes
 - Upload notebook of assignment (includes several tasks)
 - **Assure that the notebooks are saved correctly**

Details of the Examination

Running of Experiments

- Template provided in Jupyter notebook
 - Includes empty spaces ... to be filled
 - Code for automatic download of data: **run these cells first**
 - Some additional code for visualizations: **no need to run**
 - **No need to wait for training to finish**
- Usage of Google Colaboratory encouraged:
<http://colab.research.google.com>
 - Make sure to be familiar with the environment
 - Know how to upload and download notebooks (as **.ipynb**) from Colab
- Usage of local installation possible
 - See installation instructions on next page

Details of the Examination

Local Installation

- Download and install Conda on your machine:
<http://docs.conda.io/projects/conda/en/latest/user-guide/install>
- Create new environment: `conda create -n DL python=3.10`
- Activate environment: `conda activate DL`
- Install required conda packages:
`conda install numpy scipy scikit-learn matplotlib jupyter opencv`
- Install torch packages via pip, see <https://pytorch.org/get-started/locally>:
`pip install torch torchvision torchaudio --index-url https://download.pytorch.org/whl/cu118`
- Install other packages: `pip install torchsummary soundfile tqdm`
- Test your installation: Jupyter notebook from mock exam

Exam Preparation

Running Experiments Locally

- Download Jupyter notebooks from OLAT/EPIS
 - Links will be provided in the exam
- Run from console (will open a browser tab):
`jupyter notebook [file].ipynb`
 - Use **Ctrl-C** on console to stop when finished
- Maybe setup editor to work with Jupyter notebooks
 - For example: Visual Studio Code setup:
<https://code.visualstudio.com/docs/datascience/jupyter-notebooks>